

orgmode for workbench lessons

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Lesson planning with `org` and the Carpentries workbench

To export with fenced blocks consider:

```
(defun export-as-workbench-md ()
  "Export the current Org buffer to Carpentries-style Markdown using Pandoc."
  (interactive)
  (let* ((source-file (buffer-file-name))
         (output-file (concat (file-name-sans-extension source-file) ".md")))
    ;; Enable fenced_divs to convert #+begin_name blocks to ::: name
    (pandoc-cmd (format "pandoc %s -f org -t markdown+fenced_divs -o %s"
                        (shell-quote-argument source-file)
                        (shell-quote-argument output-file))))

  (if source-file
      (progn
        (save-buffer)
        (shell-command pandoc-cmd)
        (message "Exported to %s" output-file))
      (error "Buffer is not associated with a file"))))
```

While this works for exporting to the same file name, with only the extension changed, using the `EXPORT_FILE_NAME` ends up trying to export an `orgmode` block containing the variable, so a bit more effort is needed.

```
(defun export-as-workbench-md ()
  "Save, strip the export line via grep, and pipe to Pandoc."
  (interactive)
  (save-buffer)
  (let* ((src (buffer-file-name))
         ;; Get the filename from the buffer keywords
         (kv (org-collect-keywords '("EXPORT_FILE_NAME"))))
    (out (or (cadr (assoc "EXPORT_FILE_NAME" kv))
             (concat (file-name-sans-extension src) ".md")))
    ;; Grep excludes the line -> Pipe -> Pandoc
    (cmd (format "grep -vE '^#\\"+(EXPORT_FILE_NAME|PROPERTY):' %s | pandoc -f org -t markdown-smart+fenced_divs --shift-heading-level-by=1 -o %s"
                (shell-quote-argument src)
                (shell-quote-argument out))))

  (shell-command cmd)
  (message "Exported to %s" out))
```

Where the shift heading is because the workbench prefers having `##` for all lesson headings.

Along with an additional macro for the challenge headings, so `{{{markdown(### Details)}}}` with the `begin_discussion` segments work:

```
#+MACRO: markdown @@markdown:$1@@
```

and a helper to export on save:

```
# Local Variables:  
# eval: (add-hook 'after-save-hook 'export-as-workbench-md nil t)  
# End:
```

Which together, correctly render lessons for `Rscript -e "sandpaper::serve()"` as seen here in this [Python lesson](#).